

Empirical Proof Record: VALIDATED

2026-05-24T03:32:46.534089+00:00

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Proof ID: a4c2da775e1746fc0c0050b761179823...

Created: 2026-05-24T03:32:46.534089+00:00

1. Claim

Under sustained load on a balanced text corpus, Kimera's entity target completes each cycle in under 5.00 seconds at the 95th percentile (the architectural throughput ceiling).

- **Operationalisation:** 95th-percentile of per-cycle wall-time (raw['cycle_seconds']) across the streamed batch; cycles that failed to record a cycle_seconds are excluded from the denominator
- **Threshold:** p95_cycle_wall_time_s <= 5.0 seconds
- **H0:** p95 wall-time > 5.00s
- **H1:** p95 wall-time <= 5.00s

2. Verdict

VALIDATED — observed 4.197375525604002

measured per-cycle wall-time on 40/40 cycles (0 adapter errors); distribution: median 1.47s, p95 4.20s, p99 6.69s (range 0.30-8.29s); batch totals: wall 90.59s, cpu 89.72s (periodic_subprocess_sampler), rss_peak 3822.3MB, max threads 93, max procs 3

3. Evidence

Pillar	Statistic	Value	95th Percentile
0.engineering.p95_wall_time	p95_cycle_wall_time_s	4.197375525604002	
0.engineering.wall_time_distribution	cycle_wall_time_distribution	1.4651527499081567	
0.engineering.batch_cpu	batch_cpu_total_s	89.722139759	
0.engineering.rss_peak	rss_peak_bytes	3822256128.0	

4. Pre-registration

- Registered at: 2026-05-24T03:31:15.714820+00:00
- Config hash: 311bd781c294b3d83c2235bf6767fba8...
- Data hash: 83109a27c3df2a45b912398b969fa8f7...

5. Signature

24b38add7a4badc11db9a9a0f24dcda8f2b968e453a7eb43cd6b99f2d7fc57d3